

1. Administrative & Study Identification

Full Study Title: A Multicentre Randomized Double-Blind Placebo Controlled Phase 3 Study to Evaluate the Efficacy and Safety of Anifrolumab in Adult Patients with Active Proliferative Lupus Nephritis **Short Title/Acronym:** IRIS **Posting ID:** 1134995191858373126 **Protocol Number:** D3466C00001 **NCT Number:** NCT05138133 **Trial Status:** Recruiting **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Anifrolumab (SAPHNELO®) **ATC Code:** L04AA51 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/saphnelo-epar-product-information_en.pdf **Phase of Development:** Phase 3 **Medical Monitor:** AstraZeneca Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of anifrolumab administered intravenously versus placebo, as an add-on to standard of care (mycophenolate mofetil [MMF] and glucocorticoids), in adult patients with active proliferative Class III or Class IV Lupus Nephritis (with or without concomitant Class V). * **Secondary Objectives:** To evaluate the achievement of complete renal response (CRR) at Week 52, sustained reduction in oral corticosteroid (OCS) dosage to ≤ 7.5 mg/day, and time to sustained CRR.

2.2 Background & Rationale Lupus nephritis (LN) is one of the most common and severe organ manifestations of systemic lupus erythematosus (SLE), occurring in up to 50% of patients and increasing the risk of dialysis and early mortality. The current standard of care—high-dose glucocorticoids plus MMF or cyclophosphamide—carries significant toxicity burdens and often fails to yield complete renal responses. Anifrolumab, a fully human monoclonal antibody that blocks the type I interferon receptor subunit 1, demonstrated encouraging efficacy and acceptable safety in the Phase 2 TULIP-LN trial, warranting this Phase 3 trial to address the substantial unmet medical need for targeted, effective, and tolerable LN therapies.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind * **Randomization:** 1:1 Randomized * **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration * Target N: 360 participants
* **Number of Sites:** Multicenter, multinational * **Estimated Study Start:** 15-Feb-2022 * **Estimated Primary Completion:** 26-Feb-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 70 years. 2. Diagnosis of active proliferative LN Class III or IV (either with or without the presence of Class V) according to the 2003 ISN/RPS classification. 3. Renal biopsy obtained within 6 months prior to signing the ICF or during the screening period. 4. Urine protein to creatinine ratio > 1 mg/mg (113.17 mg/mmol). 5. Estimated Glomerular Filtration Rate (eGFR) ≥ 35 mL/min/1.73 m² (as calculated by the Chronic Kidney Disease Epidemiology Collaboration formula). 6. Fulfills the updated 2019 EULAR/ACR SLE classification criteria. 7. No signs of symptoms of active TB prior to or during screening or no treatment for latent TB.

4.2 Exclusion Criteria 1. A diagnosis of pure Class V LN based on the recent renal biopsy obtained within 6 months prior to signing the ICF or during Screening. 2. Known history of a primary immunodeficiency, splenectomy, or any underlying condition that predisposes the participant to infection, or a positive result for HIV confirmed by the central lab at Screening. 3. Evidence of hepatitis C or active hepatitis B. 4. Any history of cancer, except successfully cured squamous or basal skin carcinoma and cervical cancer in situ. 5. Receipt of heavy use of LN therapies for the current flare, such as IV cyclophosphamide > 2 pulses of high-dose, average MMF > 2.5 g/day for more than 8 weeks, or recent exclusionary use of calcineurin inhibitors (e.g., tacrolimus, voclosporin, cyclosporine) or belimumab. 6. Known intolerance to ≤ 1.0 g/day of MMF. 7. Previous receipt of > 2 investigational treatments for LN or SLE. 8. Any history of severe COVID-19 infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Intravenous infusions of anifrolumab compared to matching placebo, dosed every 4 weeks. * **Route of Administration:** Intravenous (IV). * **Duration of Treatment:** The total study duration may be up to approximately 116 weeks, including the Screening and Follow-up. Participants will receive anifrolumab or matching placebo throughout the Treatment Period.

5.2 Concomitant & Prohibited Therapy * Permitted: Standard of care therapies for LN, strictly defined as mycophenolate mofetil (MMF) and systemic glucocorticoids (which undergo an intended tapering protocol). * **Prohibited:** Concurrent administration of other targeted biologic disease-modifying antirheumatic drugs (e.g., belimumab), experimental therapies, and unapproved

regimens of cyclophosphamide or calcineurin inhibitors outside the standard of care protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, Medical History, Confirmation of Renal Biopsy, Baseline Labs (eGFR, UPCR), HIV/TB Screening
Baseline	Day 1	Randomization, First IV Dose, Baseline Vital Signs and Efficacy Questionnaires
Interim	Every 4 Weeks	IV Infusion (Anifrolumab or Placebo), Safety Labs, Efficacy Assessment (UPCR, eGFR tracking), OCS Dose Adjustment
Double-Blind End	Week 76	Primary Efficacy Assessment Checkpoint
End of Study	Up to Wk 116	Final Vital Signs, Exit Interview, Safety Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Difference in proportion of participants with Complete Renal Response (CRR) at Week 52. CRR is defined as maintaining a urine protein to creatinine ratio (UPCR) of ≤ 0.5 mg/mg and an eGFR of ≥ 60 mL/min/1.73 m² (or no decrease from baseline of $\geq 20\%$). * **Measurement Method:** Central Laboratory Analysis of urine collections (UPCR) and serum creatinine levels.

7.2 Secondary & Exploratory Endpoints * Difference in proportion of participants achieving sustained reduction in oral corticosteroid (OCS) dosage. * Hazard Ratio (HR) of achieving sustained Complete Renal Response (CRR) in anifrolumab compared with placebo group. * Difference in the mean standardized AUC for UPCR between anifrolumab and placebo participants.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard continuous AE tracking. Special focus is directed toward infections (including herpes zoster and urinary tract infections), infusion-related reactions, and cytopenias, which are known risks associated with anifrolumab based on Phase 2 trials.
 - **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of investigator awareness.
 - **Data Monitoring Committee (DMC):** An independent DMC oversees trial safety and unblinded data at scheduled intervals.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) Set for all primary and secondary efficacy evaluations; Safety Set for all participants who receive at least one dose of the investigational product.
 - **Sample Size Justification:** Approximately 360 participants randomized 1:1 are required to provide adequate statistical power (typically $>80\%$, $\alpha = 0.05$) to detect a clinically meaningful difference in the primary endpoint of Complete Renal Response (CRR) between the anifrolumab and placebo arms.
 - **Handling of Missing Data:** Missing data will be handled via standard imputation methods for continuous variables and Non-Responder Imputation (NRI) for categorical responder endpoints (e.g., patients dropping out early are counted as non-responders for CRR).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Scheduled onsite and remote monitoring to verify protocol compliance, source data verification, and drug accountability.
 - **Audit Trail:** eCRF (Electronic Case Report Form) tracking with secure, time-stamped audit trails and data clarification processes.
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11. Study Identifiers & Governance

- **Posting ID:** 1134995191858373126
 - **Primary Protocol ID:** D3466C00001
 - **Registry Number:** NCT05138133
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Trial Status:** Recruiting
 - **Study Title:** Phase 3 Study of Anifrolumab in Adult Patients With Active Proliferative Lupus Nephritis
 - **Official Regulatory Title:** A Multicentre Randomized Double-Blind Placebo Controlled Phase 3 Study to Evaluate the Efficacy and Safety of Anifrolumab in Adult Patients with Active Proliferative Lupus Nephritis
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12. Clinical Framework (Lupus Nephritis Specific)

- **Primary Condition:** Active Proliferative Lupus Nephritis (Class III or IV)

- **Diagnosis Threshold:** Confirmed via renal biopsy within 6 months, UPCR ≥ 1 mg/mg, eGFR ≤ 35 mL/min/1.73 m²
- **Medical Methodology:** Addition of targeted Type I interferon receptor blockade to Standard of Care
- **Optimization Target:** Complete Renal Response (CRR) and OCS tapering

13. Study Procedures & Methodology

- **Management Pathway:** Standard clinical pathway for Lupus Nephritis, adding monthly IV biologic therapy.
- **Education Protocol (HCP):** Investigator protocol training, specifically on maintaining double-blind parameters and adhering to strict protocol-defined OCS tapering regimens.
- **Education Protocol (Patient):** Education on signs of LN flares, infection risk management (e.g., herpes zoster symptoms), and trial adherence.
- **Quality Audit Mechanism:** Centralized laboratory data tracking and continuous clinical data review by the sponsor.

14. Observation & Follow-Up Schedule

- **Total Duration:** Up to approximately 116 weeks (including the screening and safety follow-up).
- **Onsite Visit Cadence:** Every 4 weeks for IV administration and monitoring.
- **Remote Monitoring Frequency:** Supplemental calls/monitoring as defined by local site protocols.
- **Checklist Review Interval:** Routine eCRF data sweeps and query generation every 4-8 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]
Statistician	[Name]	_____	[DD-MMM-YYYY]